

Chapter 8—Gene Expression: The Flow of Genetic Information from DNA via RNA to Protein

Fill in the Blank

1. The usual flow of genetic information within a cell or organism is often called the " _____ " of Molecular Biology.
Ans: Central Dogma
Difficulty: 1
2. There are _____ total possible codons. Of these, the codon _____ is recognized by a ribosome as the initiator codon.
Ans: 64; AUG
Difficulty: 3
3. There are _____ stop (nonsense) codons recognized by ribosomes. The sequences of these codons are _____, _____, and _____.
Ans: 3; UAA, UAG, UGA
Difficulty: 3
4. _____ is the process of making RNA from a DNA template, where _____ is the process of making protein by reading the mRNA code.
Ans: transcription; translation
Difficulty: 2
5. A _____ is a change in the DNA sequence of an organism. A _____ is a change in the DNA that does not change the structure of the protein product.
Ans: mutation; silent mutation
Difficulty: 3
6. Splicing of eukaryotic RNA molecules removes the _____ and links together the _____ within the genetic sequence.
Ans: introns; exons
Difficulty: 2
7. _____ is the small roundworm that is the first organism of its kind to be completely mapped genetically.
Ans: *Caenorhabditis elegans*
Difficulty: 3
8. If a base is inserted into or deleted from a DNA sequence, a _____ mutation will occur.
Ans: frameshift
Difficulty: 2

9. A ribosome has two "slots" where codons and tRNA molecules fit. The names for these areas within the ribosome/mRNA/tRNA complex forms are the _____ site and the _____ site.
Ans: peptidyl; aminoacyl
Difficulty: 4
10. _____ mutations usually cause a change in the amino acid sequence by changing the "meaning" of a specific codon.
Ans: missense
Difficulty: 2
11. Processing of eukaryotic RNA usually adds a _____ cap and a _____ tail.
Ans: methyl-group; poly-A
Difficulty: 4
12. The three steps of transcription (in order) are _____, _____, and _____.
Ans: initiation; elongation; termination
Difficulty: 3
13. _____ splicing of RNA molecules can lead to the variability needed in the genetic sequences for antibody molecules to allow for them to recognize a vast number of antigens.
Ans: alternative
Difficulty: 4
14. The enzyme _____ "recharges" tRNA molecules, while the enzyme _____ catalyzes the formation of peptide bonds between amino acids during translation.
Ans: aminoacyl-tRNA synthetase; peptidyl transferase
Difficulty: 4
15. Reverse transcription is a genetic process that is unique to HIV and other viruses that are grouped in the _____ family.
Ans: retrovirus
Difficulty: 2

Multiple Choice

16. In the usual flow of genetic information (often called the Central Dogma), which of the following best represents the expression of genes?

- A) Protein - RNA - DNA
- B) DNA - RNA - Protein
- C) RNA - DNA - Protein
- D) DNA -Protein - RNA

Ans: B

Difficulty: 1

17. The cellular organelle responsible for protein synthesis is the:

- A) nucleus.
- B) mitochondria.
- C) centrioles.
- D) ribosome.

Ans: D

Difficulty: 2

18. The scientists credited with postulating the basis for the Genetic Code are:

- A) Hershey and Chase.
- B) Griffith and Avery.
- C) Watson and Crick.
- D) Cohen and Boyer.

Ans: C

Difficulty: 3

19. A codon is:

- A) a three base sequence of mRNA that codes for an amino acid.
- B) a three base sequence of rRNA that codes for an amino acid.
- C) a three base sequence of tRNA that codes for an amino acid.
- D) a three base sequence of DNA that codes for an amino acid.

Ans: A

Difficulty: 1

20. Which of these is NOT a character of the genetic code?

- A) There are the same number of codons as there are amino acids.
- B) The code is used by nearly every living organism.
- C) Some amino acids have multiple codons.
- D) The code is degenerate.

Ans: A

Difficulty: 2

21. Charles Yanofsky helped decipher the genetic code working with the _____ biosynthesis genes in _____.
- A) leucine; *Salmonella enteritidis*
 - B) phenylalanine; *Klebsiella pneumoniae*
 - C) tryptophan; *Escherichia coli*
 - D) glycine; *Serratia marcescens*
- Ans: C
Difficulty: 3
22. A sequence of DNA that reads: 5' ATGCCTGAATCAGCTTTA 3' should code for _____ amino acids after all steps of conversion into protein are complete.
- A) 5
 - B) 6
 - C) 7
 - D) 8
- Ans: B
Difficulty: 4
23. How many DIFFERENT amino acids could be coded for using the synthetic mRNA sequence of (5' UGCUGCUGC 3')?
- A) 0
 - B) 1
 - C) 2
 - D) 3
- Ans: D
Difficulty: 4
24. There are _____ usually found in the genetic code.
- A) 3 start codons and 1 stop codon
 - B) 2 start codons and 2 stop codons
 - C) 1 start codon and 3 stop codons
 - D) 0 start codons and 4 stop codons
- Ans: C
Difficulty: 2
25. The pattern or sequence in which a molecule of mRNA is deciphered by a ribosome is called the:
- A) synthesis platform.
 - B) code degeneration.
 - C) peptide transition.
 - D) reading frame.
- Ans: D
Difficulty: 3

26. Which of these terms is NOT used as a nickname for a stop codon?

- A) emerald
- B) amber
- C) opal
- D) ochre

Ans: A

Difficulty: 3

27. Nonsense codons are:

- A) codons that code for multiple amino acids.
- B) codons that code for no amino acids.
- C) codons that can be read forward or backward.
- D) start codons.

Ans: B

Difficulty: 2

28. Which of these is NOT a step in transcription?

- A) replication
- B) initiation
- C) termination
- D) elongation

Ans: A

Difficulty: 3

29. Splicing of transcripts normally occurs only in:

- A) both prokaryotes and eukaryotes.
- B) only prokaryotes.
- C) only eukaryotes.
- D) mitochondria, as they have their own DNA.

Ans: C

Difficulty: 1

30. The sequences within mRNA that are spliced out (removed) are called:

- A) extremes.
- B) exons.
- C) inclusions.
- D) introns.

Ans: D

Difficulty: 1

31. The (expressed) sequences within mRNA that code for gene products are called:

- A) extremes.
- B) exons.
- C) inclusions.
- D) introns.

Ans: B

Difficulty: 1

32. The retroviruses, including HIV, are unique because they do:

- A) reverse replication.
- B) reverse transcription.
- C) reverse splicing.
- D) reverse translation.

Ans: B

Difficulty: 2

33. A typical example of a genetic system that employs RNA splicing is:

- A) the genes that code for the segments in an earthworm.
- B) the genes responsible for making antibodies in humans.
- C) the HIV virus genes.
- D) the genes that enable RNA splicing.

Ans: B

Difficulty: 4

34. Experiments that are done outside a living cell are describe by the Latin term:

- A) *in vino*.
- B) *in vito*.
- C) *in vivo*.
- D) *in vitro*.

Ans: D

Difficulty: 2

35. Experiments that are done inside a living cell are describe by the Latin term:

- A) *in vino*.
- B) *in vito*.
- C) *in vivo*.
- D) *in vitro*.

Ans: C

Difficulty: 2

36. The enzyme that makes RNA from a DNA template is:
A) RNA-dependent DNA polymerase.
B) DNA-dependent DNA polymerase.
C) DNA-dependent RNA polymerase.
D) RNA-dependent RNA polymerase.
Ans: C
Difficulty: 3
37. Transcription occurs in the _____ and translation occurs in the _____ of EUKARYOTIC cells.
A) nucleus; cytoplasm
B) cytoplasm; nucleus
C) nucleus; nucleus
D) cytoplasm; cytoplasm
Ans: A
Difficulty: 3
38. Transcription occurs in the _____ and translation occurs in the _____ of PROKARYOTIC cells.
A) nucleus; cytoplasm
B) cytoplasm; nucleus
C) nucleus; nucleus
D) cytoplasm; cytoplasm
Ans: D
Difficulty: 3
39. Which of the following statements is true under ordinary conditions?
A) DNA and RNA last only for a short time within a cell.
B) DNA lasts forever within a cell while RNA only lasts a short time.
C) RNA lasts forever within a cell while DNA only lasts a short time.
D) RNA and DNA both last forever within a cell.
Ans: B
Difficulty: 3
40. In the modification of eukaryotic mRNA, a “cap” consisting of a/an _____ and a tail consisting of _____ are usually added to the transcript.
A) acetyl group; multiple cytosines
B) multiple guanines; methyl group
C) multiple thymines; acetyl group
D) methyl group; multiple adenines
Ans: D
Difficulty: 3

41. RNA polymerase binds to a sequence called a/an _____ before beginning transcription.

- A) promoter
- B) operator
- C) structural gene
- D) replication origin

Ans: A

Difficulty: 2

42. Ribosomes are composed of:

- A) tRNA and protein.
- B) mRNA and protein.
- C) rRNA and protein.
- D) only proteins folded together.

Ans: C

Difficulty: 2

43. A tRNA molecule looks like a compact ____ when it is folded into its functional configuration.

- A) G
- B) L
- C) C
- D) P

Ans: B

Difficulty: 1

44. Aminoacyl-tRNA Synthetase is the enzyme that:

- A) folds tRNA molecules into their proper configuration.
- B) causes tRNA molecules to bind to the aminoacyl site of a ribosome.
- C) produces tRNA by “reading” DNA molecules.
- D) adds the appropriate amino acid to an uncharged tRNA.

Ans: D

Difficulty: 3

45. Which enzyme forms peptide bonds between adjacent amino acids during polypeptide synthesis?

- A) RNA polymerase
- B) phosphotransferase
- C) peptidyl transferase
- D) ribonuclease

Ans: C

Difficulty: 2

46. A bacterial (prokaryotic) ribosome is composed of _____ subunits.

- A) 20S & 40S
- B) 30S & 50S
- C) 40S & 60S
- D) 50S & 70S

Ans: B

Difficulty: 3

47. An anticodon is a physical component of a/an _____ molecule.

- A) tRNA
- B) mRNA
- C) rRNA
- D) DNA

Ans: A

Difficulty: 1

48. Which of these processes are coupled (linked) in prokaryotes but NOT in eukaryotes?

- A) transcription and translation
- B) replication and transcription
- C) replication and translation
- D) replication, transcription, and translation are not separate in prokaryotes

Ans: A

Difficulty: 2

49. A mutation that is characterized by a change in the DNA sequence, but no change in the resulting protein sequence, is called a:

- A) frameshift mutation.
- B) missense mutation.
- C) silent mutation.
- D) nonsense mutation.

Ans: C

Difficulty: 2

50. A mutation that changes a codon sequence, and subsequently changes the amino acid that should have been placed at that point in the polypeptide chain, is called a:

- A) frameshift mutation.
- B) missense mutation.
- C) silent mutation.
- D) nonsense mutation.

Ans: B

Difficulty: 2

51. A mutation that changes a codon that originally coded for an amino acid into a stop codon is called a:
A) frameshift mutation.
B) missense mutation.
C) silent mutation.
D) nonsense mutation.
Ans: D
Difficulty: 2
52. A mutation that occurs when a base is inserted into or deleted from a DNA sequence, completely altering an entire amino acid sequence, is called a:
A) frameshift mutation.
B) missense mutation.
C) silent mutation.
D) nonsense mutation.
Ans: A
Difficulty: 2
53. Drugs like AZT, ddC, or ddI work against the AIDS virus because they:
A) destroy viral proteins.
B) prevent the formation of viral mRNA.
C) block the action of viral tRNA.
D) look like nitrogenous bases and block viral reverse transcriptase.
Ans: D
Difficulty: 3
54. A tRNA that recognizes a nonsense codon, and inserts an amino acid where protein synthesis should have stopped, is called a:
A) nonsense tRNA.
B) suppressor tRNA.
C) revertant tRNA.
D) excision tRNA.
Ans: B
Difficulty: 3
55. Protein synthesis begins with the amino acid _____, as its codon is the one used as the “start codon” by nearly all organisms.
A) methionine
B) arginine
C) cysteine
D) leucine
Ans: A
Difficulty: 2

Matching

Match the following terms with the descriptions that best fit (each term will be used once)

- a. process of copying a DNA template strand into a complementary RNA strand
- b. nucleotide sequence that carries a complementary copy of DNA to the cytoplasm for protein synthesis
- c. cellular organelle that does protein synthesis
- d. set of codons used by living organisms (and their corresponding amino acids)
- e. molecule that carries specific amino acids for incorporation into growing protein chains
- f. protein synthesis (conversion of the RNA code into protein)
- g. set of three RNA bases that code for a specific amino acid
- h. orientation used to read the codons in the proper order for protein synthesis
- i. molecule of DNA that is copied to make RNA
- j. set of three RNA bases that code for NO amino acid
- k. set of three RNA bases that serve as the start signal for protein synthesis
- l. enzyme that makes RNA by "copying" DNA
- m. sequence of DNA nucleotides that serve as the recognition and binding site for RNA polymerase
- n. making DNA from an RNA template
- o. Eukaryotic DNA sequence within a gene that is NOT expressed in the final protein sequence
- p. Eukaryotic DNA sequence that is fully expressed in a final protein sequence
- q. unit of RNA that is removed during processing of mRNA molecules
- r. set of three bases on a tRNA that allows it to bind to mRNA
- s. molecule that serves as a physical constituent of ribosome subunits
- t. process of adding caps or tails to eukaryotic mRNA molecules

56. ____ Anticodon

Ans: r

Difficulty: 2

57. ____ Codon

Ans: g

Difficulty: 2

58. ____ Exon

Ans: p

Difficulty: 1

59. ____ Genetic Code

Ans: d

Difficulty: 3

60. ____ Initiation Codon

Ans: k

Difficulty: 3

61. ____ Intron

Ans: o

Difficulty: 1

62. ____ Messenger RNA
Ans: b
Difficulty: 2
63. ____ Nonsense Codon
Ans: j
Difficulty: 3
64. ____ Post-translational Modification
Ans: t
Difficulty: 3
65. ____ Promoter
Ans: m
Difficulty: 3
66. ____ Reading Frame
Ans: h
Difficulty: 3
67. ____ Reverse Transcription
Ans: n
Difficulty: 2
68. ____ Ribosomal RNA
Ans: s
Difficulty: 2
69. ____ Ribosome
Ans: c
Difficulty: 2
70. ____ RNA Polymerase
Ans: l
Difficulty: 1
71. ____ Spliceosome
Ans: q
Difficulty: 2
72. ____ Template Strand
Ans: i
Difficulty: 3

73. ____ Transcription

Ans: a

Difficulty: 1

74. ____ Transfer RNA

Ans: e

Difficulty: 2

75. ____ Translation

Ans: f

Difficulty: 1

True or False

76. The restoration of gene function by one mutation canceling another in the same gene is known as intragenic suppression.

Ans: True

Difficulty: 2

77. Frameshift mutations occur when the number of bases inserted or deleted is not a multiple of three.

Ans: True

Difficulty: 1

78. *In vitro* translation allows the determination of protein sequence.

Ans: False

Difficulty: 1

79. The template and RNA-like DNA strand are equivalent.

Ans: False

Difficulty: 1

80. Nonsense codon and stop codon are synonymous.

Ans: True

Difficulty: 2

81. The genetic code is absolutely universal.

Ans: False

Difficulty: 2

82. Except in reverse transcription, RNA is synthesized in a 3' to 5' fashion.

Ans: False

Difficulty: 2

83. Capping enzyme and methyl transferases are needed to add a 5' cap on mRNA.
Ans: True
Difficulty: 1
84. Exons are excised from mRNA.
Ans: False
Difficulty: 1
85. In most splice donor sites a G U dinucleotide sequence is flanked by a couple of 5' and 3' purine bases.
Ans: True
Difficulty: 2
86. rRNA carries an anticodon which is the complementary sequence to the mRNA codon that specifies a particular amino acid.
Ans: False
Difficulty: 2

Short Answer

87. The Genetic Code is described as being nearly universal, degenerate, and nonoverlapping. Briefly describe what each of these properties refers to.
Ans: The code is nearly universal...Nearly all organisms use the same codons to specify the amino acids found to correspond to the mRNA sequence. To date, only a few organisms (such as Archaeobacteria) and some organelles (such as mitochondria) DO NOT use the codons the same way.
The code is degenerate...Some amino acids are coded for by more than one codon. Though there are 64 possible codons, only 61 are used for coding purposes. There are only 20 amino acids found in living organisms' proteins, so there are many more codons than are necessary.
The code is nonoverlapping...Each codon can code for only one amino acid. For example, if a synthetic mRNA sequence of 5' UUUUUU 3' is added to the end of a transcript, only the amino acid phenylalanine will be added for those codons, showing that 5' UUU 3' codes only for phenylalanine.
Difficulty: 3

88. Describe the process of splicing in eukaryotic RNA molecules.

Ans: RNA processing is unique to eukaryotes. When the original transcript molecule is synthesized, several non-expressed sequences, called "introns," must be removed and the expressed sequences, or "exons," must be linked together in order before the transcript can be used as an mRNA. Ribonucleases make cuts at the opposing ends of introns, the "spliceosome" (intron unit) is removed. The 3' end of the first exon is linked to the 5' end of the second exon. This process is done along the entire sequence of the original transcript until all introns have been removed.

Difficulty: 2

89. Briefly describe the events of transcription in a living cell.

Ans: The DNA helix surrounding the location of a gene is unwound, and the strands of DNA are separated. RNA polymerase binds to the promoter region of the DNA, and shortly downstream from the promoter, polymerase begins matching complementary RNA nucleotides to the template sequence of the DNA. RNA synthesis continues in the 5' to 3' direction until a termination sequence is reached. At that point, RNA polymerase dissociates from the DNA, the newly-synthesized RNA is released, and the DNA helix reforms.

Difficulty: 3

90. Outline the steps of translation in prokaryotes.

Ans: The 30S subunit of a ribosome binds to the ribosome binding site of a messenger RNA molecule. A 50S subunit then joins with the other parts to complete the ribosome-mRNA complex. The ribosome "scans" the mRNA for the first 5' AUG 3' initiator codon. When the first AUG is found, a tRNA molecule matches its anticodon to the initiator codon, and fills in the peptidyl (P) site within the ribosome. A second tRNA brings in the next amino acid, filling the aminoacyl (A) site of the inside of the ribosome. The enzyme peptidyl transferase catalyzes the formation of a peptide bond between the amino acids. tRNA #1 is released to the cytoplasm, and the ribosome shifts one codon downstream. A third tRNA brings in the next amino acid, and the steps repeat themselves until all codons have been read. When the ribosome finds a non-sense codon, the entire complex dissociates and the newly-synthesized protein is free to perform its function.

Difficulty: 4

91. Describe how the "wobble" theory applies to protein synthesis.
Ans: In the genetic code, all but two amino acids have multiple codons that code for their placement into proteins through translation. Though there are 61 possible coding codons, organisms will not have 61 tRNA molecules. Some tRNA molecules will effectively bind to codons even though their anticodon is not a perfect match. This "wobble effect" usually happens at the 3' position of a codon. An example is the insertion of phenylalanine into a polypeptide. Both codons UUU and UUC code for phenylalanine. If a single tRNA is able to bind to the codon UUPyrimidine, only one tRNA would be needed to bind for inserting phenylalanine into a growing polypeptide.
Difficulty: 3
92. How does compartmentalization of transcription and translation lead to differences between prokaryotes and eukaryotes?
Ans: Eukaryotes have a nucleus and therefore transcription and translation are separated. In prokaryotes, translation occurs concurrent with transcription.
Difficulty: 2
93. Why are the genes for rRNA, tRNA and snRNAs transcribed but not translated?
Ans: These molecules are RNA species not proteins so these genes do not need to be translated.
Difficulty: 2
94. Describe nonsense suppression.
Ans: A mutation in a tRNA gene allows for an amino acid to be incorporated into a growing polypeptide rather than stopping at a stop codon created by a mutation. Therefore, one mutation suppresses the other (the mutant stop is over-ridden).
Difficulty: 2
95. What does reverse transcriptase do?
Ans: Reverse transcriptase is an enzyme that "transcribes" RNA into DNA.
Difficulty: 2
96. What is alternative splicing and how does it occur?
Ans: Introns are removed from eukaryotic mRNAs and exons are spliced together. In some genes, splicing is regulated and under certain circumstances some splicing signals are ignored resulting in changes to the primary transcript and ultimately the production of more than one polypeptide from a given gene.
Difficulty: 3

Experimental Design and Interpretation of Data

97. You are a graduate student working for a mentor whose research is top secret. You are asked to complete *in vitro* translation experiments using top-secret cell cultures to determine a genetic code for what you believe are cells isolated from extraterrestrial life. Given that you have determined that the nucleotides in the DNA are the same as on earth but the cells contain 10 unique amino acids, describe how you would definitively determine the genetic code of the “extraterrestrial” cells.

Ans: Mutation analysis would determine size of a codon and the code itself could be cracked by using short, synthetic RNA molecules in an *in vitro* translation system. A single radioactive amino acid would be added to a tRNA so that each amino acid could be tested separately. The tRNA with charged radiolabeled amino acid, plus a single “codon” would be mixed with lysate containing ribosomes. The mixture would then be poured through a filter where the ribosome + tRNA would be too large to pass through. Whenever the filter contains radioactivity, that codon codes for the radiolabeled amino acid being tested. You need to test all ten amino acids for all combinations of codons to determine the complete genetic code.

Difficulty: 4

98. What are silent, missense and nonsense mutations? Compared to control wild-type samples how would each mutation affect the outcome of Northern and Western analyses?

Ans: Silent mutations are gene mutations that have no consequences at the phenotypic level. A missense mutation changes a codon so that it codes for a different amino acid while a nonsense mutation results in the conversion of a “coding” codon to a stop codon or vice versa. The size of the bands on a Northern blot would not change (a change in stop for translation does not change the size of the RNA). It is unlikely that intensity would change either unless the coding region had some unknown effect on regulation. On a Western blot, silent and missense mutations would show no change in size however, a nonsense mutation would likely either increase or decrease the size of the protein.

Difficulty: 3

99. Hypothetically speaking, a particular protein named GOTEAM is 1,000 amino acids and is known to be involved in the phenotype, “avid football fan.” Some individuals have been genotyped and were found to have a deletion of 2 nucleotides near the 5' end of the *GOTEAM* gene. Other individuals have been found to have a deletion of 99 bases in the same region. What is the length of *GOTEAM* gene?_____ Which mutation do you think would be more likely to be the most deleterious to the GOTEAM phenotype?

Ans: The coding region of *GOTEAM* is 3,000 base pairs in length. The 2 bp deletion would most likely be more deleterious because it will change the reading frame of the gene while the 99 bp deletion is a multiple of 3 and the protein will essentially be missing a 33 amino acid region...depending on its involvement in the function of the protein, it could be unimportant or relatively important.

Difficulty: 4

100. In your studies of an unusual prokaryotic organism, you have cloned and sequenced an interesting gene and the protein that it codes. Although you have sequenced the gene several times, the start codon is always read as GTG instead of ATG. How do you interpret this result?

Ans: The prokaryotic organism from which you cloned your gene has an alternative “start” site; GTG instead of ATG.

Difficulty: 3